

PNsAV

Probabilistic Neuro-symbolic Argument Validation Model

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Abstract. PNsAV is a multi-layer neuro-symbolic reasoning system designed to perform validation of argument structures extracted from natural language. A constrained Large Language Model orchestration is employed as a semantic parser, identifying argumentative units, inferential relations, and domain concepts. These structured arguments are processed by a set of formal validation layers, used for checking the structural integrity as well as the presence of cycles and graph density. We integrate a logic engine that takes into account each argument’s internal structure through Spaans’s intrinsic strength, which is determined by numeric strengths assigned to premises and rules manually according to a specified rubric, and the global relationships between arguments, to compute a converging final strength for every argument.

1 Introduction

In recent years, the rapid advancement of Large Language Models, and their deployment in reasoning environments outpaced our ability to formally validate their outputs. While the output capacity of these models is substantial, in critical fields such as law or medicine, LLMs can generate factually inconsistent content, a phenomenon termed "hallucination" (*I*), without a traceable reasoning path, thereby challenging the safety of autonomous decision-making.

Formal logic presents multiple frameworks and tools that aid proving factual information. The current methods of mapping arguments and debates lack the scalability of a neural system, as most of them are to be manually populated with data.

Current literature explores the bridge between the two, as shown in Alfano’s 2026 FAKB pipeline (14), or Pan’s 2023 Logic-LM (12), but they often fail to target the internal structure of the arguments, remaining focused on the global acceptability, being divergent in most cases from the actual strength of the arguments themselves.

We propose PNsAV, a multi-layer end-to-end neuro-symbolic reasoning system that couples a constrained Large Language Model orchestration for natural language parsing, reducing the risk of hallucinations, with a symbolic layer of validation and logical pipeline capable of assessing both local and global structure in order to derive competent strength values starting from a manual premise and rule assessment.

2 Related Work

The development of a complete neuro-symbolic architecture capable of extracting and validating argumentative structures requires extensive research in formal logic and in natural language processing and AI. The suite of metrics for scoring reasoning proposed by Olga G. et al. in 2023 (1) proves that LLMs can exhibit specific behaviors such as incoherence, redundancy, or hallucinations. However, as identified by Miller, Nate (2026) (2), a multi-agent orchestration can be an efficient and cost-effective argument mining pipeline.

In order to categorize, analyze and express arguments mathematically, we use specific frameworks. First proposed by P. M. Dung in 1995 (3), the Abstract Argumentation Framework established the theoretical foundation for many others, such as the ASPIC⁺ framework for structured argumentation (4). The Abstract Argumentation Framework is structured on the relations between the arguments, represented as abstract nodes in a graph. Utilizing this structure, a system can compute the extension, a set of arguments which are logically justified, under a variety of formal defense semantics. ASPIC⁺ is a framework designed to augment Dung’s AAFs, offering information on the nature of the attacks and defeats as well as on the structures of the arguments, to improve the modeling of reasoning problems.

Beyond the qualitative in/out/undecided status of an argument, weighted bipolar argumentation graphs (15) assign each argument a graded strength in $[0, 1]$, propagated through attack and support relations, allowing for more fine-grained reasoning about argumentative disputes. However, these

frameworks typically leave an argument’s initial strength as an exogenous, externally supplied value. Spaans (19) addresses this gap by studying principled methods for deriving an argument’s intrinsic strength directly from the weights of the premises and rules used to construct it, rather than treating it as atomic. Separately, convergence of strength propagation on cyclic graphs remained poorly understood until Mossakowski and Neuhaus (16) introduced a modular decomposition of aggregation and influence functions and gave the first convergence and divergence results for cyclic graphs, later simplified and generalized by Potyka (18) via the contraction mapping principle, at the cost of a trade-off with open-mindedness.

Merging those two methodologies, the neural argument mining and the symbolic formal logic computations, emerges an end-to-end neuro-symbolic system that maximizes the scalability and validity of both methods. Pre-LLM neuro-symbolic systems showcase scalability issues arising from the structural dependency on problem-specific modules (5–10). Newer neuro-symbolic models, that solve the natural language parsing using LLMs, such as Logic-LM (12) and SymbCoT (11), still either lack defeasible logic (12), or use an LLM only architecture (11). Latest literature, like the FAKB pipeline by Alfano et al. in 2026 (14), presents structural flaws, more specifically in the way that they denote an argument’s strength. In the cited paper, the strength is derived from the parser LLM’s log probabilities. That causes a structural misalignment because the confidence score outputted by the LLM is based on the extraction precision rather than the argument’s actual potential value.

While alternative neuro-symbolic integration strategies are actively being explored to stabilize generative AI outputs, PNsAV uniquely focuses on an end-to-end pipeline that maps unstructured dialogue directly into weighted argumentative structures that are compositionally analyzed to provide intrinsic strength values that can be used as a basis for evaluating argumentative disputes.

3 Theoretical Basis

3.1 Dung’s Abstract Argumentation Framework (3)

Let $\mathcal{AF} = \langle \mathcal{A}, at \rangle$ be a pair where $\mathcal{A}$ is a set of arguments and at a binary relation on $\mathcal{A}$, $at \subseteq \mathcal{A} \times \mathcal{A}$. Therefore, we say that $at(A, B)$ represents an attack from argument A to-

wards argument B . Let $\mathcal{L} : \mathcal{A} \rightarrow \{\text{in}, \text{out}, \text{undecided}\}$ be a labeling function (13). We define $at_{\mathcal{AF}}(A)$ as $\{B \mid at(B, A)\}$, where $A, B \in \mathcal{A}$. Let $in(\mathcal{L})$ be defined as $in(\mathcal{L}) = \{A \in \mathcal{A} \mid \mathcal{L}(A) = \text{in}\}$ and $out(\mathcal{L}), undecided(\mathcal{L})$ to be defined analogously. $\mathcal{L}$ is *admissible and complete* iff $\forall A \in in(\mathcal{L}), at_{\mathcal{AF}}(A) \subseteq out(\mathcal{L})$ and $\forall A \in out(\mathcal{L})$, there is B such that $B \in in(\mathcal{L})$ and $at(B, A)$ holds and $\forall A \in undecided(\mathcal{L}), \nexists B \in at_{\mathcal{AF}}(A), \mathcal{L}(B) = \text{in} \wedge \exists B' \in at_{\mathcal{AF}}(A), \mathcal{L}(B') \neq \text{out}$. Note that since $\mathcal{L}$ partitions $\mathcal{A}$ into exactly three sets, each direction of the standard bi-conditional follows from the other two conditions by exclusion. By adding external constraints we arrive at different semantics. Some common ones include Grounded Semantics, where $in(\mathcal{L})$ is minimal or Preferred Semantics where $in(\mathcal{L})$ is maximal.

3.2 ASPIC⁺ (4)

An ASPIC⁺ argumentation system is built over a logical language $\mathcal{L}$, closed under negation, together with a knowledge base $\mathcal{K} \subseteq \mathcal{L}$, partitioned into axiom premises $\mathcal{K}_n$, which are certain and not open to attack, and ordinary premises $\mathcal{K}_p$, which are defeasible. Inference is governed by a set of strict rules $\mathcal{R}_s$ of the form $\varphi_1, \dots, \varphi_n \rightarrow \psi$ and defeasible rules $\mathcal{R}_d$ of the form $\varphi_1, \dots, \varphi_n \Rightarrow \psi$.

An argument A over $\langle \mathcal{K}, \mathcal{R}_s \cup \mathcal{R}_d \rangle$ is constructed recursively: (i) if $\varphi \in \mathcal{K}$, then $A = \varphi$ is an argument with conclusion φ , premises $Prem(A) = \{\varphi\}$, and no subarguments; (ii) if $A_1, \dots, A_n$ are arguments with conclusions $\varphi_1, \dots, \varphi_n$ and $\varphi_1, \dots, \varphi_n \rightarrow \psi \in \mathcal{R}_s$ (resp. $\Rightarrow \psi \in \mathcal{R}_d$), then $A = A_1, \dots, A_n \rightarrow \psi$ (resp. $\Rightarrow \psi$) is an argument with conclusion ψ , premises $Prem(A) = \bigcup_i Prem(A_i)$, and immediate subarguments $Sub(A) = \{A_1, \dots, A_n\}$.

Arguments interact through three canonical attack types. An argument B *undermines* A if the conclusion of B negates an ordinary premise used in A ; B *rebuts* A if the conclusion of B negates the conclusion of a defeasible-rule application in A ; and B *undercuts* A if the conclusion of B negates the applicability of a defeasible rule used in A . Given the resulting set of arguments $\mathcal{A}$ and attack relation $at \subseteq \mathcal{A} \times \mathcal{A}$, the pair $\langle \mathcal{A}, at \rangle$ is precisely a Dung abstract argumentation framework in the sense of Section 3.1, and the labeling semantics established there apply to it directly.

3.3 Weighted Argumentation

To support quantitative reasoning over the structures introduced in Section 3.2, we adopt a weighted argumentation graph formalism, adapted from the weighted bipolar approach of (15) by restricting it to a single, unweighted attack relation. A *WAF* is a tuple $\Gamma = \langle \mathcal{A}, R, \tau \rangle$, where $\mathcal{A}$ is a set of arguments, $R \subseteq \mathcal{A} \times \mathcal{A}$ is an attack relation, and $\tau : \mathcal{A} \rightarrow [0, 1]$ assigns each argument an initial strength, with 0 denoting total rejection, 1 total acceptance, and intermediate values expressing degrees of acceptance. Following Mossakowski and Neuhaus (16), we treat the computation of final strength as *modular*: an aggregation function α combines the strength of an argument's attackers, and an influence function ι maps this aggregate, together with the argument's initial strength, to an updated strength. This separation is what allows the propagation mechanism in Section 3.5 to be stated independently of how initial weights are obtained – in our case, from the compositional intrinsic strength of Section 3.4, rather than an atomic value. We adopt a semi-weighted formulation: only arguments carry numeric weight, and all attack types contribute equally to α , except undermining, which is excluded outright since its associated doubt is already absorbed into the attacked premise's intrinsic strength. Independently weighting attacks by type or severity is left as future work.

3.4 Intrinsic Strength

Standard WAF semantics leave the initial weight $\tau(a)$ of an argument unexplained, treating it as an exogenous input. Spaans (19) addresses this by studying principled methods for deriving an argument's *intrinsic strength* directly from the weights of the premises and inference rules used to construct it, rather than assigning it as an atomic value. We adopt this approach, using the premise weights $w(\varphi)$ and defeasible rule weights $w(r)$. The intrinsic strength $\sigma(A)$ of an argument A is defined recursively over its structure: if A is a premise argument, $\sigma(A) = w(\varphi)$ for its constituent premise φ ; if A is constructed by applying a rule r to subarguments $A_1, \dots, A_n$, then

$$\sigma(A) = \min(w(r), \sigma(A_1), \dots, \sigma(A_n)).$$

This weakest-link aggregation follows the possibilistic tradition (19) of treating an argument as no stronger than its least reliable component, and satisfies the monotonicity principles required by

Spaans’s framework. Unlike approaches that derive initial strength from a model’s extraction confidence, $\sigma(A)$ is grounded exclusively in the rubric-assigned plausibility of A ’s premises and rules, avoiding the conflation between parsing reliability and evidential strength discussed in Section 2.

3.5 Gradual Semantics

To compute final acceptability degrees over arbitrary, potentially cyclic argument graphs, we require a propagation mechanism whose convergence does not depend on acyclicity. Discrete successive-update schemes, and continuous models such as the Quadratic Energy Model (17), guarantee convergence for acyclic graphs but offer, at best, empirical evidence of convergence on cyclic ones. We instead adopt the modular semantics framework of Mossakowski and Neuhaus (16), extended by Potyka (18), which derives convergence guarantees on arbitrary graphs, cyclic included, from the contraction mapping principle: if the influence function ι has a bounded derivative (Lipschitz constant strictly less than 1), the resulting update procedure is guaranteed to converge to a unique fixed point regardless of the graph’s cyclic structure. We select an influence function satisfying this bound, using $\sigma(A)$ (Section 3.4) as the initial weight in place of an atomic strength value. As noted by Potyka (18), convergence guarantees obtained this way trade off against *open-mindedness*, the extent to which final strength can move away from the initial weight, which we account for when selecting the Lipschitz bound.

4 Structural Overview

Figure 1 details PNsAV’s internal architecture, separating the pipeline into 4 distinct modules, interconnected by bidirectional data routes. The UI/UX layers acts as an intuitive interface for users, while also being the input/output module. The LLM extractor layer, represented by a ternary LLM orchestration, is tasked to parse the natural language into conventional ASPIC⁺ structure. Structural verification and graph data are executed by the symbolic validation layer, while the actual logic is computed by the engine, the final data being routed back to the user interface.

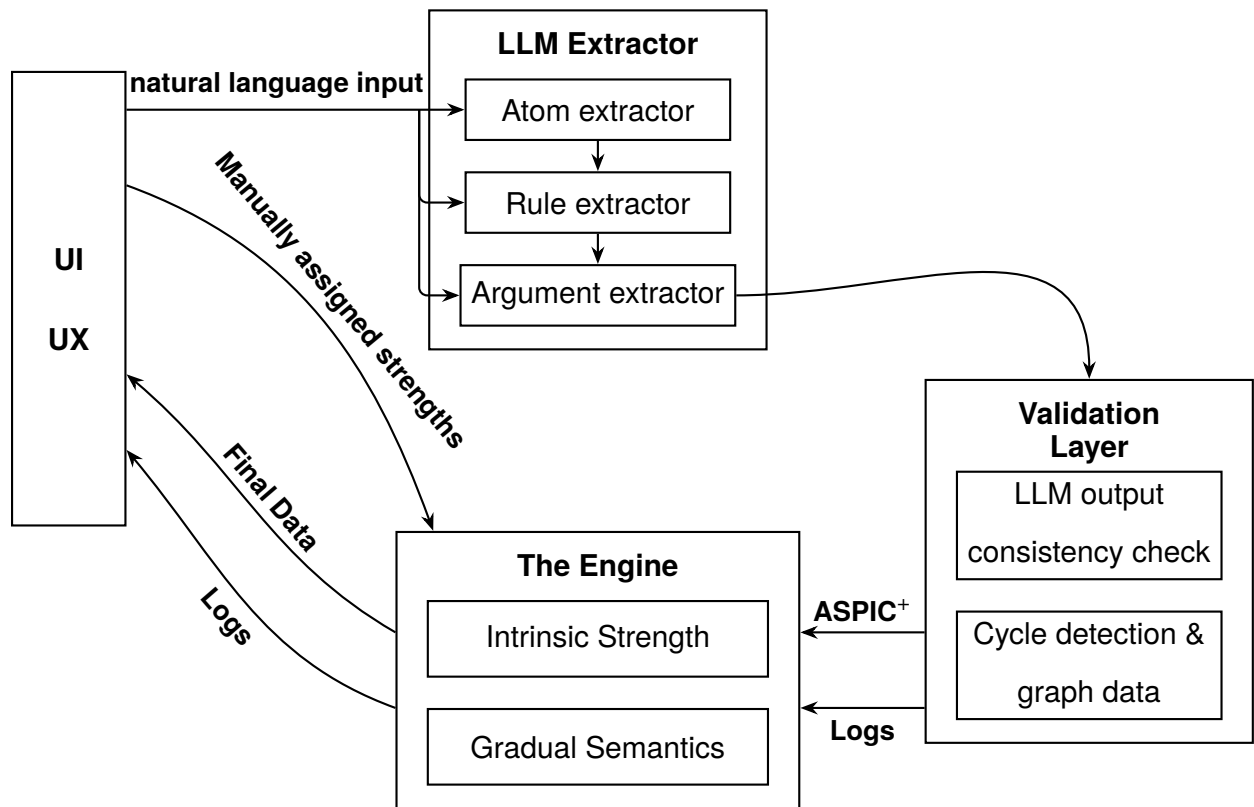


Figure 1: PNsAV's pipeline

5 Implementation

5.1 UI/UX

The PNsAV interface is fully integrated with the Python backend, using a multi-page setup built with a responsive two-column layout (`st.columns([1.5, 3.5])`) and a maximized (`layout="wide"`) screen view. The left panel handles data entry, allowing users to upload documents through (`st.file_uploader`) or type text directly into an (`st.text_area`). When you click the primary “Analyze” button (`st.button`), the app saves the data using (`st.session_state`) and starts processing it in the background. The right panel then shows the live results. The top section features an interactive graph built with `streamlit-agraph`, which renders arguments as colored nodes and connections as directed edges using a physics simulation. Directly below, the “Analyzed Text” section applies matching HTML highlights to ensure that key arguments share the exact same color codes as their corresponding nodes in the graph. At the same time, the “Logs” window updates automatically, using (`datetime.now().strftime`) to print the exact time and status dot next to every step the program takes. To give the app a professional look, we replaced the standard Streamlit design with custom CSS elements injected directly into the code. This allowed us to build a custom dark theme, rounded borders, specific box heights, and clean spacing optimized for long analysis sessions. Finally, web navigation is handled by a top bar split into columns: three buttons (“Resources”, “About”, and “Contact”) that let users switch pages by updating the session state, while the “PNsAV” logo on the left corner acts as a home button to instantly bring the user back to the main page.

5.2 Argument Mining

Natural language input is processed through a three-stage constrained LLM pipeline, structurally aligned with the ASPIC⁺ argument construction process introduced in Section 3.2: an *atom extractor* identifies premise-level propositions (annotated as axiomatic or normal premise, a *rule extractor* identifies inferential relations between them (annotated as strict or defeasible), and an *argument extractor* composes the resulting atoms and rules into structured arguments. Each stage is implemented as an independently prompted, guideline- and few-shot-constrained call, rather than a

single end-to-end extraction pass. This decomposition allows each stage’s output to be validated and constrained before being consumed by the next, isolating extraction errors to a specific stage rather than routing unconstrained free-text output directly into argument construction. Unlike approaches that fine-tune dedicated extraction models (e.g., (14)), this pipeline requires no labeled argument-mining dataset, at the cost of relying on prompt-level rather than parameter-level constraints on extraction quality. To account for this, we use chain-of-thought prompting (11).

5.3 The Validation Layer

PNsAV ensures that the extracted logic is not prone to crashing the logic layer by utilizing the Validation Layer. This layer unpacks the JSON provided by the LLM Extractor, which is confined in a strict schema, and verifies that the atoms, rules and arguments are constructed correctly (eg. an argument cannot have a conclusion different than the conclusion of its top rule). Using a basic DFS algorithm, the code checks for cycles, outputting a log to the user, displaying the graph type.

5.4 The Engine

The engine is the implementation of the Intrinsic Strength (19) and Gradual Semantics (16) explained in sections 3.4 and 3.5. The atoms, rules, arguments and attacks are given by the Validation Layer, and after that finishes running, the user is prompted with providing the strength values for all the defeasible premises and rules. Using all of the gathered data, the engine applies the propagation algorithm. The user introduces a κ scaling coefficient, and either a ε convergence threshold or a number of iterations, i . The engine then iteratively applies the following function on all of the arguments, until the values converge, or reaches the number of iterations:

$$\rho(a) = \text{clip}(\sigma(a) + \kappa \cdot \tanh(\alpha), 0, 1)$$

where $\rho(a)$ is the computed final strength, $\sigma(a)$ is the fixed intrinsic strength, and α is the net aggregated attack pressure on a . Since the influence function $\kappa \cdot \tanh(\cdot)$ has a bounded derivative (Lipschitz constant strictly less than 1), convergence to a unique fixed point is guaranteed regardless of the graph’s cyclic structure (18).

5.5 Technical Aspects

PNsAV uses Python and C++, combining ease of use and access to modern APIs with the extended performance that C++ offers. Extraction and engine execution are decoupled into independent subprocesses communicating via structured JSON, allowing each stage to be tested, rerun, or fail independently without compromising the rest of the pipeline. The dominant cost in the overall pipeline is LLM API latency during the extraction stage rather than the symbolic reasoning layer. The system was developed and tested using Python 3.13 and a MinGW-w64 C++17 toolchain on Windows, although cross-platform support is in a working state.

6 Conclusions and future work

We presented PNsAV, a neuro-symbolic system for validating natural-language argument structures by combining LLM-based extraction with formal argumentation semantics. Our central contribution addresses a construct-validity gap identified in prior neuro-symbolic argumentation systems: rather than deriving an argument’s strength from a language model’s extraction confidence, which measures parsing reliability, not evidential plausibility, PNsAV computes each argument’s intrinsic strength compositionally from the rubric-assigned strength of its constituent premises and rules, following Spaans’s principled aggregation framework. This intrinsic strength is then propagated through a gradual semantics whose influence function is constructed to satisfy a bounded-derivative condition, guaranteeing convergence to a unique fixed point regardless of the underlying attack graph’s cyclic structure, a property not established for the propagation mechanisms used in comparable prior systems.

Several limitations are acknowledged explicitly rather than left implicit. Premise and rule strengths are currently assigned manually via a rubric-constrained procedure rather than estimated automatically; while this decouples strength estimation from strength composition and avoids the plausibility conflation identified in prior work, a validated automated alternative remains an open problem. Finally, the convergence guarantees relied upon are inherited from prior work on modular bipolar argumentation semantics rather than proven fresh for our specific composition of intrinsic strength with gradual propagation; while no violation of these guarantees was observed across our test cases, a formal verification of this specific composition remains future work.

These limitations point directly to future directions: automated, construct-valid premise strength estimation; extension to a fully-weighted framework with independently weighted attack relations; explicit undercutting via rule-applicability gating; and formal verification that the substitution of compositional intrinsic strength for an atomic initial weight preserves the convergence guarantees of the underlying propagation semantics. More broadly, we see PN_sAV as a step toward argumentation systems in which an LLM’s fluency is held accountable to a formal, traceable structure, one in which an argument’s strength reflects the credibility of what it is built on, not merely the confidence with which it was parsed.

References and Notes

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